

Linux: set date through command line

Asked 12 years ago Modified 1 year, 6 months ago Viewed 769k times



How to change the system date in Linux ?

134



I want to change:



- Only Year
- Only Month
- Only Date
- Any combination of above three

linux date clock

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edited Aug 22, 2014 at 21:58



Gilles 'SO- stop being evil'

872k ● 207 ● 1.8k ● 2.3k

asked Aug 22, 2014 at 9:40



SatishWagh

15.5k ● 16 ● 72 ● 105

6 You can use e.g. `date --set='-2 years'` to set the clock back two years, leaving all other elements identical. You can change month and day of month the same way. I haven't checked what happens if that calculation results in a datetime that doesn't actually exist, e.g. during a DST switchover, but the behaviour ought to be identical to the usual "set both date and time to concrete values" behaviour. – [Christian Severin](#) Sep 29, 2017 at 9:47

Assuming you're trying to set the date to the current time, you could do `sudo ntpd -gq` to have the system update automatically using the ntp service. – [Steven Soroka](#) Mar 23, 2018 at 21:20

7 Answers

Sorted by: Highest score (default) ▾



Use `date -s`:

176

`date -s '2014-12-25 12:34:56'`

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the year/month/day is more of a challenge and there are also GUI date tools built in to the major desktop

and substitution in the date string:

will change the date, but keep the time. See [man date](#) for formatting details to construct other combinations: the individual components are `%Y`, `%m`, `%d`, `%H`, `%M`, and `%S`.

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edited Aug 22, 2014 at 10:01

answered Aug 22, 2014 at 9:44



Michael Homer

79.2k 17 224 240

I don't want to change the time – [SatishWagh](#) Aug 22, 2014 at 9:51

7 There's no option to do that. You can use `date -s "2014-12-25 $(date +%H:%M:%S)"` to change the date and reuse the current time, though. – [Michael Homer](#) Aug 22, 2014 at 9:55

came here and hours as well as minutes were matching current time. – nice – [Alex2php](#) Mar 30, 2018 at 10:36

3 `date: cannot set date: Invalid argument` – [Chris Stryczynski](#) Aug 2, 2018 at 16:17

@MichaelHomer @SHW it is possible to change only the date with a command like `date -s 2018-01-01`.

Someone shared this in an answer also: [superuser.com/questions/870068/...](#) – [baptx](#) Oct 4, 2018 at 14:15

System time

You can use `date` to set the system date. The GNU implementation of `date` (as found on most non-embedded Linux-based systems) accepts many different formats to set the time, here a few examples:

set only the year:

```
date -s 'next year'
date -s 'last year'
```

set only the month:

```
date -s 'last month'
date -s 'next month'
```

set only the day:

```
date -s 'next day'
date -s 'tomorrow'
date -s 'last day'
date -s 'yesterday'
date -s 'friday'
```

set all together:

```
date -s 1999-03-13-11:31:30 #set the year, month, day, timestamp
```

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it with the hardware clock:

```
hwclock --show
```

You can set the hardware clock to the current system time:

```
hwclock --systohc
```

Or the system time to the hardware clock

```
hwclock --hctosys
```

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edited Feb 12, 2025 at 13:09

answered Aug 22, 2014 at 9:59



Rilwan Smith

91 ● 7



chaos

49.5k ● 11 ● 129 ● 148

- 4 Bonus points for illustrating the non-obvious ways you can set the date with examples and for pointing out the difference between the "system time" and the "hardware clock". Very helpful answer! – [Mark Edington](#) Oct 6, 2016 at 2:17

I think you confused hctosys and systohc. hctosys means "Set system time from hardware clock" and systohc means "Set hardware clock from system time". – [itaych](#) Apr 12, 2022 at 9:45

▲ The command to *change the system date* is `date`.

12 There are two ways to [call the date command](#)(in Linux):

```
date [OPTION]... [+FORMAT]
date [-u|--utc|--universal] [MMDDhhmm[[CC]YY][.ss]]
```



Easy

The easiest way is to use `date -s` as it allows the use of *simple relative* dates

```
$ date -s yesterday; date
date: cannot set date: Operation not permitted
Sat Jan  5 07:21:07 EST 2019
Sun Jan  6 07:21:07 EST 2019
```

The date did not change because it was executed with a limited user `$`. If you actually want the date changed, use root (`#`) or sudo:

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aming it:

```
$ date -s "+6 months"
Sat Jul  6 08:28:39 EDT 2019

$ date -s "+3 hours -13 minutes"
Sun Jan  6 11:16:59 AST 2019
```

Absolute dates are a bit more complex as they need more detail:

```
$ date -s "2001-07-23 10:11:12"
```

Or, you can use the date command twice:

```
$ date -s "$(date +%Y-%m-%d %H:%M:%S)"
```

replace any of the `%` by a valid value and the date will be set (only as root).

```
$ date -s "$(date +%Y-11-%d %H:%M:%S)"
Wed Nov  6 08:37:15 EST 2019
```

direct

The second date call form is used to **directly** change the system date.

```
date [-u|--utc|--universal] [MMDDhhmm[[CC]YY][.ss]]
```

So :

```
date 11230812          # MMDDhhmm
```

Will set the date to the 23th of November at 08h and 12min.

[BSD has a similar command](#) but a different format ([[[[cc]yy]mm]dd]HH]MM[.ss]).

Try `date` as a limited user to see what it would do (without changing anything):

```
$ date 11230812
date: cannot set date: Operation not permitted
Sat Nov 23 08:12:00 EST 2019
```

Or, if you actually want to change the date, as root:

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And, if restarted will reset the date back to the real

```
$ date 1123081222
date: cannot set date: Operation not permitted
Wed Nov 23 08:12:00 EST 2022
```

Or a CCYY to set year **and** century:

```
$ date 112308121982
date: cannot set date: Operation not permitted
Tue Nov 23 08:12:00 EST 1982
```

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edited Jul 6, 2019 at 0:28

answered Jan 6, 2019 at 12:42



user232326



6



You change the date with the `date` command. However, the command expects a full date as the argument:

```
# date -s "20141022 09:45"
Wed Oct 22 09:45:00 BST 2014
```

To change part of the date, output the current date with the date part that you want to change as a string and all others as date formatting variables. Then pass that to the `date -s` command to set it:

```
# date -s "$(date +%Y12%d %H:%M)"
Mon Dec 22 10:55:03 GMT 2014
```

changes the month to the 12th month - December.

The date formats are:

- `%Y` - Year
- `%m` - Month
- `%d` - Day
- `%H` - Hour
- `%M` - Minute

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edited Aug 22, 2014 at 10:34

answered Aug 22, 2014 at 9:57



garethTheRed

35.3k 4 102 107

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2015-09-30 10:05:59.990" Wed Sep 30 10:05:59 +0530
y 23, 2019 at 9:58

e:

```
sudo timedatectl set-ntp 0
sudo timedatectl set-time "2022-12-03 $(date +%H:%M:%S)"
```

When date modification is no longer necessary, you should enable NTP sync again to auto update system date:

```
sudo timedatectl set-ntp 1
```

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answered Dec 5, 2022 at 4:54



1111161171159459134

1,291 ● 1 ● 9 ● 4

This applies only to installations that use systemd. Also, once you turn off ntp, you can still just use the `date` command to set the date if needed (depending on the purpose). – [éclairevoyant](#) Feb 17, 2023 at 23:07

2 This saved me some time while debugging issues in Ubuntu 22.04. Thank you! – [Machado](#) Mar 20, 2023 at 18:48

For ones like me running ESXI 5.1, here's what the system answered me

```
~ # date -s "2016-03-23 09:56:00"
date: invalid date '2016-03-23 09:56:00'
```

I had to uses a specific ESX command instead :

```
esxcli system time set -y 2016 -M 03 -d 23 -H 10 -m 05 -s 00
```

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edited Apr 16, 2019 at 18:07

answered Mar 23, 2016 at 9:10



Rui F Ribeiro

58.2k ● 29 ● 158 ● 242



Balmipour

199 ● 6

5 Thanks for the -1, but would you mind explaining *why* ? If you got a better/more complete/more exact answer, feel free to comment. – [Balmipour](#) Jul 6, 2017 at 13:34

I used the date command and time format listed below to successfully set the date from the terminal shell command performed on Android Things which uses the Linux Kernal.

```
date 092615002017.00
```

```
MMDDHHMMYYYY.SS
```

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Brook Oldre

1 ● 1

- 1 Since this does not work with either GNU or BSD `date`, could you please add more info about what system this is relevant for and how it's relevant to this particular question. – [Kusalananda](#) ♦ Sep 26, 2017 at 21:07



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